

Mill is a 1st touch

10 The Nineteenth Century: The Authority of Science

Exam question (based claim, how would you respond)

A Note on the Debt to the Nineteenth Century

Contemporary psychology, in its broadest features, remains a nineteenth-century enterprise. This is by no means to say that modern psychology is “old-fashioned” or behind the times. Still, it must be noted that the problems that consume the energy of the contemporary psychologist were either set forth explicitly in the nineteenth century or were introduced by those whose educational and cultural backgrounds were provided by the unique perspective of the nineteenth century. Of those who can lay claim to a significant contribution to the manner in which the contemporary psychologist advances the discipline, only B. F. Skinner (1904–1989) was born in this century. All the rest—Lashley, Piaget, Freud, Adler, Hull, Pavlov, Tolman, Köhler, Watson, Dewey, and James—are products of the century now under consideration. This fact is not a mere fact, but a suggestive and a revealing one. Modern psychology is not “modern” in the sense that modern physics and modern biology are. Recent discoveries of the molecular biology of the gene have transformed genetics into a discipline that Mendel scarcely would recognize, and the general theory of relativity has required the contemporary physicist to view the Newtonian universe through Einsteinian lenses.

In psychology, the situation is really quite different. The full range of its theoretical problems—from the study of personality and child development to inquiries into the neurophysiological basis of emotion or language to attempts to comprehend the determinants of social and national movements—can be traced directly to the thoughts and experiments of the psychologists and “natural philosophers” of the nineteenth century. Physicists no longer test the validity of Ohm’s law or seek to discover whether, in fact, Maxwell’s displacement current exists. They do not commit their lives to repeating those experiments that confirm the law of the conservation of energy or of angular momentum. Nor do they seek the “aether,” or insist that light must be either particulate

or wave-mechanical. Physics still has problems, very substantial problems, and some are rooted in the nineteenth century. But it is not nineteenth-century physics that is brought to bear upon these problems now. The same may be said of chemistry and the more developed branches of biology. But in contemporary psychology, not only have the problems of the nineteenth century survived, but so have many of the methods developed in that century. More important than even this fact is that the contemporary perspective is largely the one bequeathed by the scholars of that time. This will be made clearer in the next chapter. For now, we need only remain mindful, as we explore the psychological endeavors of nineteenth-century philosophers and psychologists, that our subject is only partly historical.

The Legacy of the *Philosophes*

The great British philosophers who followed in Locke's path had a decidedly scientific orientation. They were all admirers of Newton and they all either recommended the new "experimental philosophy" or actually engaged in it. Some, such as Hartley, went the route of biological psychology; others, like J. S. Mill, pursued the philosophy of science. In France, too, the disciples of Locke and Gassendi included philosophers of a seriously scientific nature, addressing their energies to specific, experimental questions. In the abiding tension between nature and spirit, they stood foursquare behind naturalism. But France, in the second half of the eighteenth century, hosted more than a group of industrious scientists and thoughtful philosophers. It hosted the *philosophes* of the Enlightenment, who had a greater effect upon the manner in which the average citizens of Paris viewed themselves and their world than did the writings of any "respectable" philosopher. The intellectual groundwork for the French Revolution was not accomplished by Descartes, less by Locke and Newton. Rather, the groundwork was done by men and women who were literary people, not philosophers or scientists. It was done by dramatists, lawyers, and, as they called themselves, *dilettantes*. The most famous members of the group were of course, Voltaire, Diderot, Rousseau, Condorcet, and D'Alembert.¹ Helvetius and Baron D'Holbach, though not members of the inner circle, derived inspiration from and voiced many of the central elements of the program of the *philosophes*. If we are to judge J. S. Mill as the nineteenth-century culmination of the empirical tradition created by Bacon, Hobbes, Locke, Newton, and Hume, then we must locate the source of Auguste Comte's *Positive Philosophy* in the French Enlightenment. Mill and Comte will occupy our attention shortly. But before turning to the specific and formalized contributions the nineteenth century made to psychology, we should have the major themes of the Enlightenment before us.

In the previous chapter, we noted that the French materialists were engaged less in science than in politics or ideology. Accordingly, the works of La Met-

trie, D'Holbach, and others had little influence on science in their own time or thereafter. The same must be said of the "encyclopedists." Neither D'Alembert nor Diderot contributed a method or a set of findings that served as a starting point for any major effort in science or in psychology. But, collectively, the *philosophes* did provide the starting point for all subsequent departures from orthodoxy. They established a liberated way of thinking and did this with such wit, craft, imagination, and penetration that defenders of the status quo inescapably invited ridicule. No single work summarizes their program nor can one be said to have launched the movement, but Voltaire's *Letters Concerning the English Nation*² came close to accomplishing both. As with a good many books of the Enlightenment, this one, too, was ordered burned by the *parlement* (1734). The *Letters* are written in an offhand way, perfectly suited to the tastes and the time of busy and influential Frenchmen. They include praise for the pioneering efforts of Descartes but an utter dismissal of his metaphysics. Newton is offered as the master, and Bacon as his harbinger. The edition of the *Letters* condemned by the *parlement* concluded with a waspish assault on Pascal's *Pensées*. Pascal's superstitions, his historical errors, his reliance on chance, his gloominess, are all pricked by the sharpest sword in Europe.

Voltaire died in 1778. For nearly fifty years, his writings and very personality were the hub from which the scholarship of the Enlightenment radiated. He was enormously wealthy, and his influence was further increased by a close friendship with Frederick the Great of Prussia, the poet-tyrant and most liberal king in Europe. Voltaire's example gave confidence to Diderot (1713–1784), whose *Encyclopédie* suffered a painful on-again, off-again experience at the hands of the *parlement*. Indeed, many of the important works banned in France saw their way to Prussian publishers through the intercession of Voltaire. In his assertion of the superiority of the Lockean-Newtonian philosophy to that of Cartesianism, Voltaire set in motion a spirit of sensualism that led to Condillac and to Helvetius. The latter (1715–1771) published *A Treatise on Man, His Intellectual Faculties and His Education*,³ which comes as close to twentieth-century environmentalism as any work written before 1900. It is in the *Treatise* that hereditary differences are dismissed as negligible and that the effects of training, reward, punishment, and experience are accorded first place in determining the character and accomplishments of the individual.

Voltaire also played a part, through Diderot, in the formation of Baron D'Holbach's materialistic philosophy, a materialism that soon took the form of polemical atheism. It was Holbach (1723–1789), as much as La Mettrie, who found the mechanistic portions of Descartes' psychology sufficient to account for morality, emotion, intellect, and language. It was Holbach who railed so vituperatively against established religion that the entire circle of encyclopedists were soon treated as atheists, en masse.

Voltaire is credited with writing some twenty thousand letters to more than a thousand different correspondents. His plays moved the masses, his ideas, the

philosophes. His, more than any other single voice in the eighteenth century, spoke the cause of freedom, reason, law, and humanist ethics. He believed in God but was not religious. He believed in science but made no contribution to the literature of science. As with Diderot and Helvetius, he had received a Jesuit education and respected, all his days, the authority of reason. Many feared that he learned his lessons too well! He was no revolutionary, nor was he a "democrat" as that term is now employed. He insisted, however, that the ultimate validity of any government is rooted in its contributions to the welfare of its citizens, an idea that would become immortalized in Rousseau's *Contrat Social*. He, the encyclopedists, the growing power of the middle class, and the conflicts between king and *parlement*, between *parlement* and church, between Jesuit and Jansenist—these were the seeds of revolution and reform. All the leading figures of nineteenth-century science and natural philosophy looked back to the scholars of the Enlightenment for support and inspiration.

We may summarize what they found when they looked back. First, the idea of progress. In the works of Voltaire and, most particularly, in the rational materialism of Diderot and Condorcet, we discover repeatedly the notion of personal and cultural evolution. It is in *D'Alembert's Dream*⁴ that Diderot presents the view of the whole as a collection of material parts, of the statue gaining its life from material reduction and subsequent evolution. And Condorcet (1743–1794), in his *Sketch for a Historical Picture of the Progress of the Human Mind*,⁵ written as its author hid from the vengeful zealots of the Revolution—that is, from those whose new freedoms Condorcet had labored to secure—set to paper the idea that powered the entirety of the nineteenth century: the idea of progress.

Second, the idea of nature. If Voltaire, Diderot, D'Alembert, Condorcet, D'Holbach, Rousseau, and the rest can be said to have agreed on any single point—and the disagreements among the group were considerable—that point is philosophical naturalism. The world and everything in it are matter. The world is to be comprehended as matter in motion. Human reason, by which this comprehension becomes possible, must aim itself at nature and unearth nature's laws. To the Pascals who insist that we can never know everything, Voltaire's reply rings down the decades to our own day:

Let us console ourselves for not knowing the possible connections between a spider and the rings of Saturn, and continue to examine what is within our reach.⁶

Included in the idea of nature was the idea of natural law as applying to all spheres of reality. It was in this same period that Turgot and the *physiocrats* (*physis* = nature; *krateo* = power, sovereignty) argued for a "free market" economic policy whereby the "law" of supply and demand would set the "natural" value on goods and labor.

Third, the idea of personal freedom. Rousseau's greatest essay begins with the haunting picture of man, *born free and everywhere in chains*. Here is the spirit of the Enlightenment, transported by an Englishman to America and translated as *The Rights of Man*. Tom Paine would even become an elected member of the post-Revolution Assembly, notwithstanding the fact that he scarcely knew a word of French.

The Legacy of Kant

Kant died at the beginning of the nineteenth century, but he is far more a member of the Age of Reason than he is the Age of Materialism. He set the tone of German philosophy for the entirety of the nineteenth century and, as a result, German "naturalism" would be ever barren of the strong empiricist elements of the naturalistic philosophies of France and England. German philosophy, that is, would come to accentuate the transcendentalism of Kant.⁷ The combination of naturalism and transcendentalism yielded that uniquely German creation, Romantic Idealism, whose principal architect was Johann Wolfgang von Goethe (1749–1832).

spirit
vill. above
world
material

It was the poetry of Goethe that animated the German-speaking world with the spirit of *Sturm und Drang*. His *Sorrows of Young Werther* (1774) resulted in an epidemic of suicides. Naturalism, in the hands of Goethe, was the panpsychism of Leibniz, Spinoza's creative forces. Man stands hopelessly alone in an immense universe, torn by the stresses of life, and finding himself only through his own activity, through his life. The goal of life is living of it, the activity and personal evolution, the reaching upward and outward. Life is love and passion. Kant's rejection of teleology was all Goethe needed to convince himself that the only end of human life is the activity that defines it. Nature is but a system of opposing forces: life and death, light and dark, love and hate.

lowers Existentialism

Empiricism in the Nineteenth Century

We concluded Chapter 7 with a discussion of the Utilitarian philosophy of Jeremy Bentham and observed that this system would be modified and promulgated by many, particularly J. S. Mill (1806–1873). We pick up the evolution of empiricism—this time as empirical psychology—with Mill.

There is no need to go into Mill's "conversion" to Benthamism and philosophical radicalism at age fifteen. His *Autobiography*⁸ is widely quoted, and the missionary labors of his father, James Mill, in behalf of Benthamism have already been extolled. Mill never abandoned the broadest features of utilitarianism, but he did recognize that Bentham's founding version of it was insufficient to meet the demands of the time. Unlike Bentham, who came to philosophy by way of law and economics, Mill's background was in classics, logic, and science. Where Bentham wrote in the grand style of the eighteenth-century

universalists, Mill was more systematic and far less intuitive. His era was far more complex than Bentham's and he appreciated the complexities. Thus, we do not find the Mill of *On Liberty* attempting to "prove" the validity of freedom in terms of physics, or mathematics, or logic. Nor do we find the Mill of the *System of Logic* defending the inductive method in terms of economic or social considerations. Mill is a distinctively nineteenth-century gentleman, sophisticated, educated, liberal, and urbane. He is, in short, a modern as we would apply the term.

J. S. Mill's most important work in psychology was his *System of Logic* (1843), which was immediately successful and, through eight editions in his own lifetime, served as the handbook of the scientific community. The bulk of the work was devoted to a description of the principles of induction, the failure of purely rational approaches to matters of fact, the methods to be employed in establishing valid inferences, and the role of deductive processes in science. Contemporary science continues to rely on Mill's "methods," and our own easy commitment to the hypothetico-deductive method can be attributed to the immediate hold the *System of Logic* took on the scientific mind. Not many scientists in the late nineteenth century read Galileo; all of them read Mill.

The portion of the *System of Logic* most important to the appearance of experimental psychology is Book VI and especially Chapters III–VIII.⁹ Having laid the foundation for all science, Mill now addresses his attention to human nature as a subject of science:

It is a common notion, or at least it is implied in many common modes of speech, that the thoughts, feelings, and actions of sentient beings are not a subject of science. . . . This notion seems to involve some confusion of ideas, which it is necessary to begin by clearing up. Any facts are fitted, in themselves, to be a subject of science, which follow one another according to constant laws; although these laws may not have been discovered, nor even be discoverable by our existing resources.¹⁰

Mill offers an illustration from meteorology. We are not able, he says, to specify all the antecedent variables to produce rain, but we all agree that rain-fall is the result of laws of nature. Thus, forecasting the weather is a probabilistic endeavor and may always be. That is, it may never be an exact science, but it is a science nonetheless.

The science of human nature is of this description. It falls far short of the standard of exactness now realized in Astronomy; but there is no reason that it should not be as much a science as . . . Astronomy.¹¹

He notes that the subject matter of psychology embraces the thoughts, feelings, and actions of human beings and that these cannot be predicted with anywhere near the accuracy as that obtaining in astronomy. This limitation, how-

ever, does not indicate that psychology cannot be a science or even that it is not a science. We cannot foresee every circumstance in which an individual might be found, and the factors that conspire to form the individual character are so diverse that even if we knew these future circumstances, we might still not be able to predict how the individual will act. Nevertheless we must still accept that actions, feelings, and thoughts are caused, and that the causes are natural and, to that extent, knowable in principle. Some of the laws are, in fact, known already, according to Mill, and central among these are the laws of association, which Mill summarizes as follows.¹²

First, there is Hume's law according to which every mental impression has a corresponding idea. Once we have experienced X, we are able to recall X without its actual presentation. We are so constituted that we are able to form an idea or mental image of that which we have perceived.

Second, there is the law of connection such that the repeated, simultaneous (or immediately successive) presentation of two stimuli leads us to think of either when the other is later presented. This law, cited by all the empiricists, is most ably described, Mill tells us, by one James Mill, whose *Analysis of the Phenomena of the Human Mind* presents the law "with a masterly hand." The third law is one that renders the intensity of a stimulus relatively interchangeable with the frequency of its presentation. According to this law, a very intense X has the same effect upon the mind as does a weaker Y that has been presented more frequently.

These simple or elementary Laws of Mind have been ascertained by the ordinary methods of experimental inquiry; nor could they have been ascertained in any other manner. But a certain number of elementary laws having thus been obtained, it is a fair subject of scientific inquiry how far those laws can be made to go in explaining the actual phenomena. It is obvious that complex laws of thought and feeling not only may, but must be generated from these simple laws.¹³

Mill's empirical psychology was not of the radical sort. He certainly looked to education and the general cultural environment for the causes of individual differences, but he was equally persuaded that organic factors might be and probably are responsible for some of the more dramatic differences. He noted the advances being made in neurophysiology and neurology and believed that in time we would have a much better understanding of the relationship between brain physiology and the laws of the mind. However, as with Locke and Hume, he refused to take a position on the material basis of thought and noted that Hartley and his own father had displayed more confidence in the materialist account than was justified by available data.¹⁴

There is a second respect in which Mill's empirical psychology was conservative. He tended to doubt, notwithstanding the fact that psychology could and would develop empirical laws, that the science of psychology would ever be

able to go beyond the given context in its attempts to predict human conduct, thought, and feeling. By an "empirical law," Mill meant a law of regularities: Y follows X or is coincident with it in this setting, and we may infer that Y will follow X or be coincident with it in any setting greatly resembling this one. However, in a very different context, we are unable to specify precisely what the relationship, if any, will be between them. Since the human condition is ever changing, since history ordains ever different contexts, the laws of psychology will be empirical and of restricted generality.¹⁵ Through the empirical laws, which allow us to predict the actual facts of conduct or thought or feeling in a given and restricted context, it is possible to deduce more general laws. These will not be merely empirical but will be exact. However, the price we pay for these exact laws is that they will not apply to facts but to tendencies. That is, from the empirical laws of association, we may inexactly predict that Henry Jones will form a stronger association between intense stimuli than he will between weak ones. There may be a race of men about whom this is not true; yet, from this "situational law," we can deduce that, if there is thought at all, associations will be formed. Note that the law of associations does not predict or even attempt to predict the *exact* result of an experiment. Rather, it refers to the tendency of something to occur, other things being equal. That other things are never equal in the affairs of man is only to say that our exact laws will not be testable, not that they are not laws. Mill gave (invented) the name "ethology" to cover this science, which is deduced from the empirical laws of psychology. As conceived by him, ethology was to be the science of character, or that discipline concerned with the effects of environmental conditions on the laws of thought, feeling, and conduct. It was to be "the Exact Science of Human Nature," whose propositions are "hypothetical only, and affirm tendencies, not facts,"¹⁶ Modern ethology, of course, bears only a slight resemblance to Mill's expectations. This is not the occasion to examine the subtleties of Mill's plan for ethology, but we must, before leaving the point, draw attention to the ambiguous distinction between that which affirms a tendency and that which affirms a fact. It is worthwhile to draw attention to Mill's choice of words here because the *philosophical* behaviorism of several twentieth-century writers (e.g., Ryle) has come to rely heavily on the notion of dispositions and tendencies as opposed to observable facts.

The very mention of behaviorism, perhaps by one of those laws of association, brings one to Mill's version of utilitarianism. Over and against Kant's categorical imperative, which Mill sees as potentially admitting "the most outrageously immoral rules of conduct,"¹⁷ Mill adopts the "happiness theory":

Questions of ultimate ends are not amenable to direct proof. Whatever can be proved to be good, must be so by being shown to be a means to something admitted to be good without proof.¹⁸

We judge medical science to be “good” because it conduces to health, but we have no way of showing that health itself is good. We accept this without proof, its ultimate sanction coming from the fact that all or very nearly all humankind will attempt to possess it. To those who reject utility as being no more than the measure of goodness employed by the beasts, Mill counters that there is nothing in utilitarianism that declares that human happiness is limited to pleasures of the flesh or bestial wants. Utilitarianism accepts the facts and the needs of the human intellect and recognizes that, for humans, the greatest happiness is not confined to mere biological gratification. Pleasures differ in quality as well as in quantity, and no utilitarian denies it:

If I am asked what I mean by difference of quality in pleasures, or what makes one pleasure more valuable than another merely as a pleasure, except its being greater in amount, there is but one possible answer. Of two pleasures, if there be one to which all or almost all who have experience of both give a decided preference, irrespective of any feeling of moral obligation to prefer it, that is the more desirable pleasure.¹⁹

Students of modern behaviorism will recognize in this passage the forerunner of that notion according to which it is the organism, not the psychologist, who defines the reinforcer. Pleasure and the quality of pleasure are to be judged in terms of what is sought by individuals when they have access to a variety of possibilities. Pain is that which is avoided by individuals when they are able to reveal their judgments in their conduct. And, lest the imaginative reader attempt to reconcile this view with some version of Kantian morality, Mill makes his own position utterly transparent:

There is, I am aware, a disposition to believe that a person who sees in moral obligation a transcendental fact, an objective reality belonging to the province of “things in themselves,” is likely to be more obedient to it than one who believes it to be entirely subjective, having its seat in human consciousness only. But whatever a person’s opinion may be on this point of ontology, the force he is really urged by is his own subjective feeling, and is exactly measured by its strength. . . . It is not necessary, for the present purpose, to decide whether the feeling of duty is innate or implanted. Assuming it to be innate, it is an open question to what objects it naturally attaches itself. . . . If there be anything innate in the matter, I see no reason why the feeling which is innate should not be that of regard to the pleasures and pains of others. If there is any principle of morals which is intuitively obligatory, I should say it must be that. If so, the intuitive ethics *would* coincide with the utilitarian.²⁰

Thus, Mill not only doubts that the intuitivist (i.e., Kantian) can ever prove his moral system but argues that even if, somehow, he is right, the utilitarian system can easily assimilate the “native” pleasures and pains. The important point, from Mill’s perspective, is to understand the source of power that moral injunctions have. That source is none other than the pleasure or pain caused by our actions or anticipated as consequences of our actions. If there is such a thing as an ultimate morality, its sanctions derive from this consideration and only from this consideration.

The connection between utilitarianism and empiricism is direct. If morality is to be a knowable subject, then it must have a factual foundation. The latter requires that the elements of any moral philosophy be observable and that its propositions be, in principle, testable. On this account, utilitarianism, according to its adherents, is the only scientific moral system. Human acts can be assessed in terms of their consequences. When these are judged by those affected by them to have produced happiness, the acts may be judged as moral. When the consequences, as judged by the actual or potential recipients, are painful, the act is immoral. All other appeals must reduce to the utilitarian. All other standards must reduce to the subjective consequences on actual people.

We have observed the conservative elements of Mill’s empirical psychology and now we must offset these with the radical character of his empirical philosophy. Unlike his forebears Locke and Hume, who were willing to exclude at least the necessary truths of mathematics from the overall realm of empirical science, Mill argued that even these truths are finally to be understood as inferences from experience. In defining matter as the “permanent possibility of sensation,” he asserted phenomenalism as the fundamental epistemology. This is not to be confused with idealism, in any of its variants, since Mill was a confirmed believer in matter. Rather, he took the tenets of epistemological empiricism to their logical limit, which indeed requires that all statements about the material world must ultimately have perceptual counterparts or referents. Thus, to the extent that mathematical propositions are propositions about the real world, they too must have their source in the data of sense. This is a philosophical and not a psychological issue and is beyond our present purpose. It is enough to indicate, however, that this feature of Mill’s empiricism rendered the entire enterprise suspect as, in many quarters, it now remains.²¹

Mill was influenced not only by the British tradition but by older contemporaries in France as well. Chief among these, and one of the most important figures in the experimentalism of the nineteenth century, was Auguste Comte (1798–1857), whose six-volume *Cours de philosophie positive* established positivism as a system of philosophy. By his own admission, Comte was led to his thinking principally by Condorcet’s *Sketch* (cited above), with its emphasis on cultural evolution and intellectual progress. And in the best French tradition, Comte labored to convert the idea into a movement that can only be called political. In outline, Comte’s position is that cultures pass through three distinct

stages: the theological, which is superstitious; the metaphysical, in which hidden physical forces or causes replace deities; and finally, the scientific, in which positive knowledge replaces the superstition and metaphysics. The three stages must occur and in the order specified. At each transition point between one and the next, the culture finds itself in a "critical period." An old perspective, which has been satisfying to the masses for an extended time, is now to be replaced by one whose virtues are only slightly sensed and only by the best minds of the period. Once the transition is complete, the culture enters an "organic" period: one of synthesis and discovery, of growth and intellectual evolution.

Much along the same lines as those provided by Condorcet in his *Sketch* of intellectual progress, Comte's positivist doctrine required each new science to emerge from the principles of an older and more established one. Each science develops methods appropriate to its problems. Illustrative is the science of sociology which Comte himself named. Its method was to be that of a comparison of cultures in various stages of evolution. And, through the discovered laws of sociology, all other sciences would be better comprehended, since all other sciences are the product of social evolution. Agreeing with Kant that the mind itself is not directly observable, and recognizing that much of what presented itself as "psychology" was no more than philosophers attempting to discover the laws of the mind introspectively, Comte dismissed psychology as "an idle fancy, and a dream, when it is not an absurdity."²² If the mind is to be studied, only two methods are available. The first, which Comte called "phrenological psychology" and which we shall discuss later in this chapter, involves research on the relation between brain processes and mental states and functions. The other is the direct observation of the *products* of mental life, and this, to Comte, was sociology. Comte believed that if the philosophical psychologists had not attempted the impossible and conceited task of looking into their own minds, and if they had, instead, appreciated the importance of *feeling* and *emotion*, they would have looked throughout the animal kingdom and discovered bona fide psychological principles. Convinced, however, that only man was rational and intelligent, they ignored this most promising investigative terrain.²³

Comte's positivism was extended to embrace religion, ethics, and economics. He and his disciples advanced a view of science that placed it on a footing no different from the world's historic religions. Science would solve all problems, answer all questions, remove all doubts. To deny its power was to sink into metaphysical or, worse, theological torpor. To avail one's self of its methods was to set the world aright. Once more, the promises of Bacon, Hobbes, Descartes, and the *philosophes* returned to haunt the orthodox believers. Now they were told that there are two kinds of statements a person can make. One refers to the objects of sense, and it is a scientific statement. The other is nonsense!

Mill, too, was a positivist, although he disagreed on several counts with

Comte's version. He (wisely) rejected phrenology, he accepted associationistic laws of the mind (even though the method of their proof might be "introspective"), and he refused to depart from strict empiricism even to the point of dogmatizing on what science *must* be. Still, his ethology was intended to complement Comte's sociology; his rules of inference were designed to yield positive knowledge; and his appeal to the senses as the ultimate arbiters of truth was unashamedly Comtean. The ideas of Mill and Comte, paced through the revisions that all formulations undergo, culminated in the logical positivism of the Vienna circle: Wittgenstein, Schlick, Carnap, Reichenbach, and their followers. It was this group that introduced the twentieth century to the most coherent and critical attack on rationalism that had been witnessed since William of Ockham. According to the logical positivists—and they might just as well be called radical empiricists—the facts of the world are *sensations*, and all the laws of science are ultimately reducible to empirical propositions. Once we have exhausted the data of sense, there is nothing else that can be said either of the world or ourselves. These developments will be explored further in the next chapter.

Darwin and the Idea of Progress

Evolution, as a fact of perception and as a literary metaphor, is a dateless concept in the human mind. Aristotle and Empedocles argued about its ontogenetic character, and Goethe extended it to the very processes of history. Condorcet, in his *Sketch*, analyzed the human experience as a series of stages beginning with tribal communities and proceeding, in orderly succession, through nine more levels, the last being scientific in language and orientation.²⁴ Comte's positivism asserted the same, more formally and in far greater detail.

If evolution has been an abiding concept, so, too, has nature-as-enemy. Every age and every person confront the challenges posed by the elements: plague, starvation, drought, illness, enfeeblement. Not only are these the ever-present features of life, but they have been perceived, since recorded history, as producing "heartier stocks," "fit specimens," and the like. The Spartan regimen recommended for Athenian youth in the *Republic* is defended on grounds such as these. The strongest metal is that tempered in the hottest flame. Even tribal initiation rites seem to respect the theory that asperity breeds future success. Darwin, then, is not to be accorded the honor of "discovering" either evolution or natural selection if by discovery we mean the same sort of process as that involved in Maxwell's discovery of the laws describing electromagnetic phenomena.

What Darwin (and, independently, Alfred Russell Wallace) discovered was that evolution, or "progress," is not only possible but inevitable and even, in a very ruthless fashion, mechanical. He and Wallace, both read and acknowl-

edged the ideas imparted by Thomas Malthus's *Essay on the Principle of Population*.²⁵ Malthus demonstrated, with mathematical rigor, that the breeding potential of the human race was staggering—but never realized. That is, it is possible for human beings to increase in number *geometrically*, but such increases do not occur. Clearly there must be an opposing force tending to limit populations. In describing what he called “the struggle for existence,” Malthus presented war, pestilence, and starvation as the forces tending to prevent geometric increases in population. According to his analysis, whenever the rate of increase in population exceeds the rate of increase in the production of food, people necessarily will die.

In the *Origin of Species*, Darwin applied Malthusian concepts to all living systems. He presented the forces of nature as blind agents of selection, “natural selection,” which either favor or reduce the probability that a species would survive. Within any species, wide (natural) variations are apparent. The press of the environment favors certain variants and these, as a result, become more numerous. Darwin (1809–1882) published his works before Mendel's findings in genetics had become generally known and, as a result, the *Origin of Species* did not contain any modern notions about the mode of transmission of the various traits involved in survival. In fact, where Darwin was specific on this count, he was wrong. But he did recognize that mutations occur, that some of these were better suited to the conditions they faced, and that, as a result, they would come to prosper, often at the expense of the parent-type. The cheerful idea of progress entertained by the *philosophes* of the Enlightenment now took on a gloomier complexion. Progress is but the consequence of annihilation. New life appears and succeeds, while older forms perish. Supporting his theory with archaeological findings, Darwin relentlessly withstood the attacks of churchmen and skeptical scientists alike. If the Bible suggests that all living forms were created at the same time, the Bible is wrong. If Christian teaching insists that God made so many forms of life and no more, Christian teaching is wrong. If philosophers believe that man, through will and accomplishments, has removed himself from the natural contexts in which survival never exceeds probability, the philosophers are wrong. Find a species of bird now in abundance and you will have the descendant of an avian form, different from the present one, which was unable to accommodate the demands of a changing environment. Man came into being as a successful variant of a primate-type no longer with us. Quite simply, he evolved.

Darwin's most important psychological work is *The Expression of the Emotions in Man and Animals*,²⁶ a book that may be said to have launched comparative psychology. Since the major strains of Darwinian theory are now widely known, we need only say a few words about his impact on psychological thought in the nineteenth century. By far, the greatest effect produced by Darwinism was that which located the psychological human on a continuum of

biological organization. In *The Expression of the Emotions*, Darwin examines the facial musculature of many species, *homo sapiens* included, and notes not only the anatomical similarities (already well established) but the similarities in facial expression produced by conditions giving rise to similar emotions. The angry dog and the actor feigning anger both retract their lips back over their teeth, bare their teeth, and clench them. Signs of submission, of sexual attraction, and of melancholy are of a similar nature throughout phylogeny wherever we find the anatomical equipment necessary for the expression of affect. Natural selection has favored those species able to form "serviceable habits." It has led to the evolution of nervous systems so constructed as to produce behaviors leading to mating, to the avoidance of tissue-damaging stimuli, to the ingestion of nutritious foods. Not only is the present species the survivor of a long process of natural selection, but in the behavior and the emotions of this species, we will discover elaborated forms of those behaviors and feelings that characterize simpler types. In quasi-teleological terms, Darwinian ethology explained variations in the living world in terms of the ultimate goal of all life: the continuation of life. Attainment of this goal requires adaptation to the exigencies of the environment. The species that cannot adapt will vanish. In evolving into a new and different form, it, itself, must cease to be. The required evolution includes not only the anatomy of the species but its functional physiology as well—its habits, reflexes, and sensitivities. Psychological evolution is, therefore, concomitant with structural, anatomical evolution. Just as we are able to discern, in the structural nuances of more advanced species, the archetypal features of earlier forms, so also can we see, in the developed psychological equipment of the advanced species, traces of more primitive dispositions and abilities.²⁷

It was not long after Darwin's *Origin of Species* that his cousin, Francis Galton (1822–1911), published his studies of "hereditary genius" (1869).²⁸ Galton had accepted the evolutionary theory without reservation and had accepted as well the epistemological empiricism that constituted England's national philosophy. Putting the two together could produce only one outcome: a theory of mental superiority based on the capacity of the senses! Galton's studies of the children of successful students indicated that mathematical ability "ran in families," although the environmentalist interpretation of the data received only a modest hearing. Galton needed only the Darwinian theory in order to explain why there is such great variability in human intelligence. He needed only an age of experimentalism and a burgeoning commercial empire to support his belief that the measurement of this variability was important. In selecting such tasks as those involving visual acuity and depth perception, Galton merely reflected one of the more innocent features of radical empiricism. In any case, a "mental measurement" tradition was begun which, lacking Malthusian restraints, has since grown geometrically.

Darwinian biology meant different things to different people, and it still

does. It was illegal to teach the subject in a number of American states until relatively recently. To religionists of a fundamentalist stripe, the whole idea was out-and-out heresy. The theory required that the earth be much older than Scripture had indicated. It argued for the continuous creation of new forms; fundamentalism insisted on a "big-bang" origin of all forms. It demanded more of man than mere virtue in the struggle for survival and it allowed the gradual elimination of man himself. At best, it traduced the Day of Judgment into a painfully drawn-out affair.

To the middle-class commercialists of Victoria's empire, Darwinism was soon interpreted as a justifying ethic. The poor were poor "by nature." Since we are all engaged in an eternal struggle against the elements, since only certain "types" are properly adapted to the demand, it is inevitable that there will be the "haves" and the "have-nots." Given Galton's findings, it is even likely that the two classes will perpetuate their respective stations hereditarily.

To the Continental scholars, so much under the influence of Comtean positivism and the "three-stage" theory of cultural evolution, the Darwinian model pulled many diverse strands of thought together. The nervous system has evolved, consciousness has evolved, structure entails function. In Freud's words, "anatomy is destiny." Simultaneously there began to appear studies of comparative culture, comparative anatomy, and, alas, comparative psychology. Everything now made so much sense: the idea of progress, the enlightened machine, utilitarianism, the positive philosophy, young Werther besieged by the *Sturm und Drang* of his own feelings. A revolution was taking place, aptly titled by one historian "heroic materialism."

Nineteenth-Century Materialism

Alexander Bain (1818–1903), a close, and admired associate of J. S. Mill, wrote the following to Mill in 1851, describing the progress he was making in his text in psychology:

I have just finished rough drafting the first division . . . which includes the Sensations, Appetites, and Instincts. All through this portion I keep up a constant reference to the material structure of the parts concerned, it being my purpose to exhaust in this division the physiological basis of mental phenomena. . . . And although I neither can, nor at the present, desire to carry Anatomical explanation into the Intellect, I think that the state of the previous part of the subject will enable Intellect and Emotion to be treated to great advantage and in a manner altogether different from anything that has hitherto appeared. There is nothing I wish more than so to unite psychology and physiology that physiologists may be made to appreciate the true ends and drift of their researches into the nervous system.²⁹

It was Bain who founded the essentially psychological journal *Mind*. It was Bain who wrote the two most influential psychology texts to appear before the twentieth century.³⁰ What is it that prompted so physiological a bias in this empiricist? Why, unlike Locke, Hume, and even Mill, did Bain anchor empirical associationism to the science of physiology? To answer these questions we must review three extraordinary developments in neurophysiology that occurred in the decades immediately preceding Bain's own contributions. These were (1) the Bell-Magendie law, (2) the law of specific nerve energies, and (3) the "science" of phrenology.

The Bell-Magendie law is named for Sir Charles Bell (1774–1842) and Francois Magendie (1783–1855), who independently discovered the anatomical separation of sensory and motor functions of the spinal cord. Bell, reporting the finding to associates at a dinner party (1811), almost lost recognition to Magendie, who presented his work in the more durable form of a published scientific article (1822). Both demonstrated the ability to render an animal insensitive by compressing or transecting the dorsal member of a spinal nerve. The animal so treated was still able to move the surgically anaesthetized part of the body but did not respond to intense stimulation delivered to that region. Similarly, by ventral-root transection it was possible to elicit cries of pain from an animal who, nevertheless, could not withdraw its limb from the pain-producing stimulus. The Bell-Magendie law was important to psychological thinking on several counts. First, it provided clear, anatomical evidence of the sort of arrangement required by the Cartesian theory of sensory-motor functioning. Perhaps most significantly, it reinforced confidence in the experimental approach to the study of sensation and behavior. While the law did not disclose the actual mechanism of reflex-formation, it provided the structural foundation on which reflex mechanisms had to be based. The contribution thus extended a line of inquiry initiated by Descartes, receiving important experimental corroboration from the work of Stephen Hales and Robert Whytt, and theoretically rich amplification by David Hartley. Bain, the most psychological of the nineteenth-century associationists, readily perceived the importance of this law to his plan for a physiological psychology.

Bell himself had anticipated a form of the law of specific nerve energies, but most of the credit for this law is owed to one of the most prominent physiologists of the century, Johannes Müller (1801–1858), whose *Handbuch der Physiologie des Menschen* (1834–1840) was the most authoritative work of the period. The law, which has become a commonplace, asserts that the quality of experience is determined not by the features of the objective stimulus but by the particular nerves responding to it. An orthodox Kantian, Müller judged the law to be only plausible; that is, we do not sense the objective world as it is but come to know about it only in the translated form delivered by the organs of sense. The "thing in itself" remains a mystery. Our knowledge is the subject-

12/17
 replication
 of
 function

diff part of brain do diff things

phrenology is a system of psychology in which the mind is divided into parts, each of which corresponds to a part of the brain (so can feel bumps & tell things about them)

fied knowledge abstracted from the objects of sense and transformed by the organs of sense. Philosophical idealism aside, however, the law of specific nerve energies placed the qualitative and quantitative aspects of experience in the nerves, that is, in *nature*. It contributed a biological feature to epistemology in that the nature of knowledge now was inextricably tied to the characteristics of the "organs of knowledge."

Where the Bell-Magendie law and the law of specific nerve energies were important additions to an emerging physiological psychology, the concept of "localization of function," as propounded and forged into a movement by Franz Joseph Gall (1758–1828), may be said to have created the discipline. We must quickly add, however, that Gall and his phrenology accelerated the appearance of physiological psychology in the sense of presenting notions that only a physiological psychology could successfully refute. Alexander Bain was much taken by phrenology, which reached the English-speaking world through J. G. Spurzheim's *The Physiognomical System of Drs. Gall and Spurzheim*³¹ and the translation of Gall's, *On the Function of the Brain and Each of Its Parts: With Observations on the Possibility of Determining the Instincts, Propensities, and Talents, or the Moral and Intellectual Dispositions of Men and Animals, by the Configuration of the Brain and Head* (1835).³² The title is presented here because it very nearly exhausts the theory contained in the work itself. It is in this work that Gall presented the four "incontestable truths" of phrenology:

[T]he brain alone has the great prerogative of being the organ of the mind. . . . The moral and intellectual dispositions are innate; their manifestation depends on organization; the brain is exclusively the organ of the mind; the brain is composed of as many particular and independent organs as there are fundamental powers of the mind.³³

It is fortunate for the history of the neural sciences that the phrenology of Gall and Spurzheim was quickly challenged by a more sober group of thinkers. Yet for the better part of thirty years it was all the rage. Numerous journals devoted to the "science" sprouted in Europe, England, and the United States. It is only in our own century that the last of these disappeared. A good part of the success of the venture may be traced to Gall's deserved reputation as a neuroanatomist and to the salesmanship of Spurzheim. Moreover, the fundamental idea behind phrenology received some support from the neurological clinic, from common sense, and from the materialistic bent of nineteenth-century psychology. Even the caveman must have recognized that the head-end of his prey was uniquely effective in rendering the animal quiescent and, if the same caveman received nonlethal blows to his own skull, this recognition was further established. Greek and Egyptian medicine and its survival through the Renaissance had catalogued a number of functions mediated by the brain. Des-

cartes, more than any previous writer, had popularized the role of the brain in experience and action. Following Decartes and the Gassendists, numberless investigators, here and there, began to disseminate their findings from neuro-surgical patients, from clumsy studies of decapitated animals, from observations of the traumatically head-injured victims of war and civil strife. La Mettrie, of course, put the whole matter on the polemical foundation needed by any movement, and several of the more medically minded *philosophes*, trailing in La Mettrie's winding path, carried both the soul and the mind confidently into the brain. D'Holbach was most prominent in this respect.

Unlike La Mettrie, Gall actually had stature as a scientist and, coming several decades after La Mettrie, had considerably more data with which to make his case. Even more important to the development of his theory than the growing number of experimental findings, however, was the firmly entrenched "faculty psychology" begun by Locke and developed by the later empirical philosophers of England and Scotland. As we discussed in Chapter 7, the idea of "faculties," as old as Aristotle's *De Anima*, received new life in the sentimental theories of seventeenth- and eighteenth-century England. By the time of Gall, it was common to accord to humans not only the "internal light" (of reason) but also the faculties of empathy, justice, love, morality, etc. Occasionally the list would become very long. Gall accepted these faculties, which is no more than to say that he acknowledged that people possessed such characteristics. Since they did, and since "the brain is exclusively the organ of the mind," it took only a slightly breathless exercise in deduction to conclude that each faculty enjoyed specific representation within the cerebral cortex.

Among those who took phrenology seriously, Bain was not the only scholar of substance. No less a figure than Herbert Spencer (1820–1903) more than toyed with the idea, and recall that Spencer was one of the earliest, most eloquent, and most successful defenders of Darwin's theories. Spencer, like Bain and Mill, was concerned that psychology take its place among the natural sciences and that it extricate itself from the purely speculative discipline of philosophy. Gall, understandably, held out the prospect of just such a liberation. In contending that morality, every bit as much as sensation and movement, was to be understood in terms of the organization and functional physiology of the nervous system, Gall gave psychologists precisely the foundation required for the development of an independent science. In addition, the broad features of phrenology did no violence to Darwinism. The latter, which emphasized instinctual drives, inherited habits, reflex mechanisms of survival, and naturally selected nervous processes, was actually an ethological or comparative phrenology, when understood a certain way. And, of course, this was the way the followers of Gall and Spurzheim understood it.

The polemical style of Spurzheim and the cultist flavor of this new science of phrenology invited immediate opposition from many quarters. The most

telling criticisms were advanced by Pierre Flourens (1794–1867), who actually tested certain phrenological hypotheses experimentally. Flourens performed surgery on animals, including removal of the cerebral hemispheres, and also carefully noted pathologic changes in the brains of recently deceased patients who, prior to death, displayed a variety of neurological symptoms. It was Flourens who insisted that the cerebral mantle functions *as a whole*, that the magnitude of a deficit is not simply reducible to the amount of brain involved, and that similar deficits can result from lesions in any of several brain regions. While Flourens was struggling to defend the vitalistic and dualistic elements of Cartesianism, he was successful, nonetheless, in drawing attention to the purely scientific deficiencies of the phrenological theory.

Gall died before Darwin's great works were written. Spurzheim carried on, defending the master's system till his own death four years later in 1832. Both lived long enough to attract the attention and the admiration of Bain and Spencer, and it was the latter two who were able to see the connection between the concept of localization of function and evolutionism. Spencer, in his very widely read *Principles of Psychology*, divorced himself from the orthodox phrenology of his own day, but then added this:

Nevertheless, it seems to me that most physiologists have not sufficiently recognized the general truth of which Phrenology is an adumbration. Whoever calmly considers the question, cannot long resist the conviction that different parts of the cerebrum must, *in some way or other*, subserve different kinds of mental action. Localization of function is the law of all organization whatever; and it would be marvellous were there here an exception. . . . Any other hypothesis seems to me, on the face of it, untenable.³⁴

Spencer, attempting to tie this to the theory of evolution, unfortunately accepted the Lamarckian notion of the inheritance of acquired characteristics and so was forced to argue that the learning, memory, and habits of a species appear in the next generation. Darwin, of course, was given to the same idea. But eliminating this error from the Spencerian system, we are left with a not-so-dated theory of mental function: the sensory fibers project to specific regions of the brain; repeated stimulation *somehow* results in a greater facility of neural transmission; chemically, prior experiences are stored within the cerebral hemispheres; and the “subjective psychology” of association is but the other side of the “objective” or neurophysiological psychology within which the associations are physical. Added to this is the Darwinian contribution according to which the complexities of human neuroanatomical organization are to be understood as the evolved forms of a more primitive organization. Thus:

The claims of Psychology to rank as a distinct science are . . . not smaller but greater than those of any other science. If its phenomena

are contemplated objectively, merely as nervo-muscular adjustments by which the higher organisms from moment to moment adapt their actions to environing co-existences and sequences, its degree of speciality, even then, entitles it to a separate place. The moment the element of feeling, or consciousness, is used to interpret nervo-muscular adjustments as thus exhibited in the living beings around, objective Psychology acquires an additional, and quite exceptional, distinction.³⁵

Psychologists getting into Psychology

Bain and Spencer did all but found experimental psychology. They fought successfully for the independent status of the science. They presented the associationistic laws in a form never fully appreciated by their predecessors in empirical philosophy, proper. They cemented the new science to the new biology and the two have never been completely divided since. Revealingly, the first chapter of Spencer's *Principles* was titled "The Nervous System," and this established a policy followed by textbook writers for nearly a century. They failed to found experimental psychology only in that they failed actually to engage in psychological research or to create the facilities with which such research could be conducted. Thus, it is to Wilhelm Wundt (1832–1920), who conducted research and who took the pains to construct and name a laboratory devoted exclusively to psychology, that the honor of "founder" is traditionally given.

Wundt prepared for psychology's first laboratory at the University at Leipzig in 1879. By that date Darwin's revolution was beginning to be generally appreciated throughout the scientific community. Bell, Magendie, Flourens, Gall, and Spurzheim had all made their varied contributions to the neural sciences. The great Hermann von Helmholtz (1821–1894) had not only written the classic works in the physiology of vision and hearing but had also measured the velocity of the neural impulse and had advanced the most compelling version of the law of the conservation of energy. The latter two contributions are worth citing in apposition since each, in its own way, challenged the remnants of eighteenth-century vitalism. In discovering that the velocity of nervous conduction was not only measurable but even rather sluggish (maximum = 120 meters/second), Helmholtz simultaneously put to rest the Cartesian view of the soul's ubiquitous influences on the body and the historic belief that mind is beyond the reach of observation. If the mind influences the body, it does so through the brain, and the conducting pathways and the brain perform their mission in measurably physical ways. Helmholtz was the most illustrious and independent of Müller's students and the most vocal opponent of the vitalistic elements in Müller's *Handbuch*. Wundt, after receiving his degree, was Helmholtz' assistant for several years and may be said to have been intimately familiar with the major neurophysiological facts and theories of his time. It is clear that no German scientist of the period escaped the physicalistic (i.e., an-

tivitalistic) climate created by Helmholtz, but it is equally clear that Wundt, in the history of psychology, was *sui generis*. Even today, one cannot read his *Principles of Physiological Psychology*³⁶ without being impressed by his sensitivity to the experimental, philosophical, and biophysical problems uniquely affecting the new science. Of his many works, this one was the most important. It was not only one of the first (and one of the few) of his books to be translated into English, it was also the most self-consciously programmatic.

With Bain and Spencer, Wundt saw the need to join the sciences of psychology and physiology. Unlike either Bain or Spencer, Wundt was a scientist—that is, a working scientist—and, as a result, he was far more concerned with the development of proper methods and measures. For these he looked to the recent discoveries of Gustav Fechner (1801–1876), also a Leipzig scientist. Fechner had published the landmark *Elemente der Psychophysik*³⁷ in 1860, the book that sought to express mathematically the relationship between mental and physical events. It was in this work that Fechner's law was set forth according to which the strength of sensation is proportional to the logarithmic value of the intensity of stimulation. In the same work Fechner had demonstrated how the properly instructed laboratory subject, paced through a series of repeated measurements, would yield reliable data reducible to lawful description. No longer was it necessary to apologize for “introspective” techniques. Comte's critique of psychology seemed to vanish under the weight of Fechner's law. The *psychophysical* methods were simple to apply and were not unlike the methods used in any quantitative science called upon to deal with variable phenomena. Wundt, seldom lavish in his praise, notes only this:

It is Fechner's service to have found and followed the true way; to have shown us how a “mathematical psychology” may, within certain limits, be realized in practice. . . . He was the first to show how Herbart's idea of an “exact psychology” might be turned to practical account.³⁸

But for Wundt, neither the rational-mathematical deductive psychology of Herbart³⁹ nor the dualism of Fechner could serve as the foundation of psychological science. Indeed, for Wundt, none of the earlier formulations was adequate. Of the British empiricistic psychology, he had this to say:

In the psychological portion of their works, these writers usually adopt the theory of the “association of ideas” elaborated in the English psychology of the eighteenth century. They adopt it for the good and sufficient reason that the doctrine of association, from David Hartley (1705–1757) down to Herbert Spencer (1820–1904), has itself for the most part attempted merely a physiological interpretation of the associative process.⁴⁰

Psychology, from Wundt's perspective, must concern itself with the "manifold of consciousness" and arrive at an understanding of far more than the laws of association. The manifold of consciousness includes the mind's commerce with more than the objects of sense, with more than external stimuli. It embraces feelings, images, dreams, memories, attention, and movement. And the psychology charged with the study of these processes is "experimental psychology," the name given by Wundt himself. The science is to be the study of mind, but by "mind," Wundt is careful to dissociate himself from that long history of metaphysical speculation:

"Mind," "intellect," "reason," "understanding," etc., are concepts . . . that existed before the advent of any scientific psychology. The fact that the naive consciousness always and everywhere points to internal experience as a special source of knowledge, may, therefore, be accepted for the moment as sufficient testimony to the rights of psychology as science. . . . "Mind," will accordingly be the subject, to which we attribute all the separate facts of internal observation as predicates. The subject itself is determined wholly and exclusively by its predicates.⁴¹

Here, in a conservative and rather woolly fashion, Wundt hands down a manifesto of Humean proportion. By "mind" the psychologist will mean no more than and only that which is directly reportable as an observation of an internal event. If the mind thinks, feels, remembers, attends, and forgets, then a science of mind can be no more than experimental inquiries into the determinants of thinking, feeling, remembering, etc. When its predicates are exhausted, there is no metaphysical residue.

For Wundt, "psychology," "experimental psychology," and "physiological psychology" were three terms for the same subject. To appreciate this, it is necessary to recall how the term "physiological" was understood by Wundt, and also to recognize the place of *social* psychology in Wundt's writing. By physiological (*physiologische*) the German scientist of the late nineteenth century described an essentially scientific, law-governed approach. The term, unlike its current meaning, did not refer specifically to biological events or measures, but to the entire range of lawful operations governing the "animal economy." This, then, is the sense in which Wundt's terms "psychology" and "physiological psychology" can be taken as synonyms. But what of "experimental psychology"? In this, as will be developed further in the next chapter, a distinction must be made between Wundt's conception of "social psychology" and our own. His later works on "folk psychology" are primarily devoted to what we call cultural and historical anthropology. It was Wundt's judgment that significant events in human history were, of course, beyond the reach of experimental methods and, more importantly, were not explicable in the language of natural science. That is, one cannot comprehend human history

“physiologically” because this history does not proceed according to exclusively causal laws. More will be said on this in Chapter 11. It is sufficient here to understand that Wundt did not take “social psychology” to be a branch of psychology, but a separate subject requiring its own methods, measures, and explanatory principles. Psychology proper, however, is synonymous with “physiological” and “experimental” psychology. Again, however, this did not commit Wundt to an exclusively *biological* psychology. He rejected out of hand that form of materialism that insisted that mind was no more than matter or the mentalist assertion that mind was irreducible to matter. He rejected, that is, the grounding of his science in metaphysical disputes. Psychology henceforth was not to be a branch of philosophy, less of biology. It was to be the experimental science devoted to an analysis of the contents of consciousness or, as he put it, the manifold of conscious experience. Ultimately this analysis would converge upon equivalent analyses of the structure-function relationships of the nervous system. Wundt took a disinterested position in the Flourens-Gall controversy, well aware that Gall spoke without having the necessary facts and that Flourens, in opposing phrenology, dismissed more of the localization theory than the facts demanded. He reviewed in his *Principles* the recently discovered technique of direct, electrical stimulation of the living animal’s cerebral cortex. Beginning in 1870, Fritsch and Hitzig published the results of their studies of the effects of direct electrical stimulation of the cerebral cortex of dogs. They demonstrated the topographic organization of sensation and movement on the cortical surface and further demonstrated the relatively separate cortical “strips” associated with sensory and motor capacities. Referring to their work, Wundt would remark on the “simplicity of the structural plan,”⁴² though the simplicity of 1875 has been seriously questioned in the intervening century. Wundt also considered the more established techniques of ablative surgery and the various methods of clinical assay. He recognized early the severe limitations of each of these methods and also foresaw that none of them was a substitute for the introspective data easily secured from a conscious, healthy human being. He had a sophisticated awareness of the fact that the mind-body problem was not to evaporate as a result of improved technology. He had, as well, a nearly modern disdain for the suggestion that it was psychology’s responsibility to settle the matter. The topics, methods, and theories that define contemporary psychology have changed considerably since Wundt’s time, but the essential attitude of the experimental psychologist today may be said to have originated in the Leipzig laboratory.

The Reflex Revisited

Even as the larger agenda of psychology was altered by research and theory drawn from neurology, more basic research in the functions of the brain and

spinal cord was reinforcing an essentially mechanistic perspective. Marshall Hall (1790–1857) was one of the leaders in these developments. He was also one of the founders of the British Medical Association and a most outspoken abolitionist, addressing this cause in his travels in the United States.

In the tradition of Whytt, Hall undertook extensive research on spinal reflexes, establishing that coordinated and integrated behavior could be elicited in the absence of any regulation or participation by higher centers. But research and clinical findings also led him to conclude that such behavior is not accompanied by consciousness. To support this conclusion he suggested the following:

[T]he cerebrum itself may be viewed as the organ of the mind on which the *psyche* sits, as it were, enthroned . . .⁴³

How different . . . are the functions which belong to the true spinal marrow! In these there is no sensation, no volition, no consciousness, nothing psychical.⁴⁴

The true spinal system never sleeps.⁴⁵

What is the hidden agent in [spinal] function? Is it *Galvanism*?⁴⁶

Hall's work was widely disseminated through Johannes Müller's influential *Handbuch*. He was also an active spokesman in behalf of his own accomplishments, claiming as much for himself as his work would allow—and perhaps a little bit more. In the process, he helped to put the concept of the reflex more securely on the map of an emerging theoretical psychology, laying the foundations for a biologically oriented behavioral science.

The "Idealist" Alternative

The Kantian legacy was briefly touched on earlier in this chapter, but the most important influences were neglected. These were the interpretations of Kantian philosophy by his immediate successors, Fichte (1762–1814), Schelling (1775–1854), and Hegel (1770–1831). Together, they forged a unique form of idealistic psychology, which still affects the manner in which Continental psychology proceeds. We neglected this facet of the Kantian legacy in part because Kant himself may not have been eager to claim it and also because the movement was fueled by those empiristic and materialistic perspectives we have just explored—fueled in the sense that this new idealism was a conscientious antagonist of empirical and physiological psychologies. Predictably, Wundt dismissed the system as no more than a "rational psychology" emerging from Kantian "nature philosophy" and having only negative effects on scientific psychology specifically and on natural science in general.

Kant's role in the appearance of German idealism is central, if miscast. Al-

categories force us to look at world in certain ways (can't help it)

hand → no scientific truth

nothing is interpreted (reason comes first) as it's perceived

only perception

though rejecting the subjective idealism of Berkeley, he described his own metaphysical position as "transcendental idealism," by which he meant to distinguish between the actual physical world of material objects and the sorts of knowledge we can ever have of those objects. The objects, for Kant, were "things in themselves" that could never be known empirically as "things in themselves." Rather, our empirical knowledge constitutes a translation of the real world, a translation performed by the pure categories of the understanding operating in conjunction with imperfect senses. To "know" is to interpret, not merely to sense, but in the very act of interpretation we suffuse objective nature with the categories. The mind that does this is not an "object" and, therefore, can never be known in the sense in which we know the natural world. This leads to the insistence, of course, that a science of psychology is scarcely imaginable. Wundt summarized the Kantian position this way:

Kant once declared that psychology was incapable of ever raising itself to the rank of an exact natural science. The reasons that he gives . . . have often been repeated in later times. In the first place, Kant says, psychology cannot become an exact science because mathematics is inapplicable to the phenomena of the internal sense; the pure internal perception, in which mental phenomena must be constructed,—time,—has but one dimension. In the second place, however, it cannot even become an experimental science, because in it the manifold of internal observation cannot be arbitrarily varied,—still less, another thinking subject be submitted to one's experiments, conformably to the end in view; moreover, the very fact of observation means alteration of the observed object.⁴⁷

We have seen that these objections did not prevent Wundt from committing his life to experimental psychology. Fechner had shown, to Wundt's satisfaction, that mathematics could, indeed, be applied to the phenomena of the "internal sense." Moreover, every branch of natural science must alter its objects in the process of observing them.

If Wundt did not perceive the Kantian objections to be telling, Fichte, Schelling, and Hegel accepted them as axiomatic. For Fichte, again in the Kantian tradition, the very freedom of the human will, in contrast to the deterministic character of purely physical processes, settled once and for all the question of a scientific psychology: there could be none. If there is to be a psychology, it must be a deductive, philosophical discipline that accepts as its subject the will and intentions of the self (*ego*). The ego affirms itself through itself and not through recourse to external objects. In affirming itself, it forces nature to conform to its will and, indeed, may even be said to animate nature through its will. While the will is free, the spiritual ideals of human life impose upon it the constraint of duty, the neglect of which is the essence of evil. In these various

Wundt
Wundt's
to external
(emphasis)
cognitive
aspect

gives
meaning
presence
(alone)
top
down
system

(argues in favor of second certain way)

elements—the self imposing its own character on nature, the transcendental nature of duty, the freedom of the will—Kant's metaphysics appears in the form of a psychology of personality.⁴⁸ Schelling's departures from Fichte's philosophical psychology require no mention here. In the significant respects, he and Fichte agreed; freedom of the will in an otherwise determined world of matter entails an inescapable dualism that no scientific psychology can eliminate. If we are to comprehend the nature of mind, our only method is that of the mind reflecting on itself and deducing the terms of its unity. Wundt, while rejecting the prescriptions of these idealists in the matter of method, could not extricate his psychology from their prescriptions regarding topic: the manifold of conscious experience.

The capstone of nineteenth-century German idealism is to be found in the philosophical and logical works of George Friedrich Hegel, whose influence on European thought is equaled, if at all, only by Descartes and Kant. Even if our present subject were the history of philosophy, it would hardly be possible to summarize "Hegelianism" in less than a very substantial chapter. Indeed, unless one is a professional philosopher, it is hazardous to attempt a summary of any sort. As with most productive scholars, Hegel underwent a change of mind from time to time. His writing suffers from the fatal combination of genius and literary awkwardness. The subjects of greatest interest to him—the Absolute, the Ineffable, Soul, Art, and Religion—tend to frustrate authors of the greatest literary skill, and Hegel is not one of these. One of the clearer expositions of his system in the English language is the old but not dated study provided by Professor W. T. Stace,⁴⁹ but even this work is so punctuated with Hegelian locutions as to leave the uninitiated . . . uninitiated. Russell surely has many advocates of his contention that Hegel is "the hardest to understand of all the great philosophers,"⁵⁰ and this from a man who believes that "almost all of Hegel's doctrines are false."⁵¹ Be that as it may, Hegel invented phenomenology, was acclaimed by some otherwise moderate Englishmen as the new Aristotle, and caused Karl Marx to describe himself as "the pupil of that mighty thinker."⁵² Hegel is not to be ignored.

Hegelian "doctrines," as Russell called them, are all derived from rational first principles. Completely rejecting Kant's caveat about the subjective nature of reason, Hegel declares that the truths of reason are necessary, nonarbitrary, and final. Through reason, the mind can dissect the apparent world such as to lay bare *its* reason. At this point, we must insert the distinction, the Hegelian distinction, between "reason and cause." Hegel, while totally opposed to the thrust of empirical philosophy, readily appreciated Hume's arguments against *necessary* causal sequences. He agreed that no logical bridge could ever be constructed whereby one could proceed from an effect back to physical causes of a necessary nature. However, while causes do not logically entail effects, reasons do. Professor Stace explains the emphasis this way:

[E]xplanation involves the idea of logical necessity. It is just the apparent absence of necessity in the world which makes us complain that it is incomprehensible. Cold produces ice. This is a fact which simply *is*. We cannot see why it *must* be. . . . If, instead of being a mere fact, we could see that it is logical necessity; if we could see the *reason* of it, and that it follows from the reason as necessarily as a logical consequent from its antecedent, then we should understand it. . . . Thus, a philosophy which would genuinely explain the world will take as its first principle, not a cause but a reason. . . . This is the fundamental Hegelian idea of explanation . . . it was for this that the Greeks, especially Aristotle, were groping, when they said that the first principle of the world is not prior to the world in time, i.e., as a cause is prior to its effect, but is *logically* prior to the world, i.e., as a logical antecedent is prior to its consequent.⁵³

Reason, then, is the first principle and it explains itself. It determines itself and the world in that, unlike causal sequences, the sequence from reason to statements about the world is directed by logical necessity. To say that John Smith is dead "because a trigger was pulled" is not to explain *why* he is dead—only to note one of the antecedent causes of his demise. We explain *why* Smith is dead when we say that "Jones wanted him dead and it was Jones who pulled the trigger." The reason is prior to the cause. Where the effect is only contingently related to the cause, it is necessarily related to the reason. Harkening back to Kant's *Categories*, we can say that the characteristic of temporal succession is *logically* tied to the a priori concept of time.

Hegel's theory of psychology is presented most clearly in his *Encyclopaedia*⁵⁴ and his *Phenomenology of Mind*.⁵⁵ There has been something of a contest among historians to come up with as many people as possible who may be said to have anticipated the theories of Sigmund Freud. We do not seek this elusive prize in recognizing Hegel's part in the development of Freud's thinking. Hegel's philosophy of mind is rife with the concepts of stages of development, ego and anti-ego conflicts, and intimations of a death wish. One could not be educated in the Austria of the 1860s and 1870s and not be influenced by the thoughts of Hegel.

The Hegelian philosophy of mind begins with the theory that the mind is a stage in the evolution of soul. Initially (and Platonically) the soul shares the realities of nature; as Stace describes this "natural soul," it is the "beginning of spirit"⁵⁶ and exists as mere being. It cannot reflect upon itself nor can it assimilate to itself the objective elements of the physical world. From this state of pure egoism, the soul, through historical development, gains "sensibility" by which it is able to distinguish between itself and its contents. While the soul still cannot comprehend *external* objects, it has nonetheless an awareness of

the difference between the feelings within it and itself. The “natural soul” is now the “feeling soul.” This stage is followed by one in which the soul actually can receive external objects, can distinguish itself from the perceptual contents resulting from sensory experience, and can catalogue the elements of the physical world according to universal categories. The soul, now the “actual soul,” recognizes itself “as its contents”; it is at one with its sensations, ideas, and feelings. Once the soul has expressed the ability to distinguish between itself and the objects of the external world, it may be said to have “consciousness” and to be “mind.” It is the study of this consciousness—as Wundt would later call it, the manifold of consciousness—that is phenomenology. Consciousness too passes through stages: “sensuous consciousness,” “sense perception,” and “intellect.” The first of these stages allows the mind to receive the “raw data” of experience, impressions devoid of cognitive features. The sensation is of an event—immediate and psychologically neutral. Perception, however, is another matter. Here, the observer adds to or brings to bear upon the merely sensuous the concept of the universal. Where the sensuous produces a “this” or a “that,” the sense perception is of the form “What is *this*?”:

Whatever we say in answer to this question invests the “this” with a universal character. . . . To say that it is “here” or “now” is at once to apply concepts, or universals, to it; for “here” and “now” are both universals. . . . Everything belongs to the class of objects which are called “this.” Hence, “this” is a *class*-name and imports a universal.⁵⁷

Sense perception, in applying the universal concept to each particular, establishes a contradiction—the contradiction between a particular object and a *class*. Perception, alone, cannot resolve the contradiction. This is the task of the intellect which, through its inventions of scientific laws and principles, recognizes the ultimate reality to be universals, and particular objects to be mere appearances or instances. Once this developed form of consciousness has succeeded in abstracting the universal principle from each of the particular sense perceptions, it recognizes that its knowledge is *idea*, for universals are, by their nature, ideas. At this point, consciousness is led to self-consciousness or the awareness of the idea of self.⁵⁸

It is hardly necessary to remind the modern reader of the “Hegelian” tone of much contemporary discourse not only within the community of professional psychologists but within the relatively nontechnical spheres of daily life. “Self-awareness,” “self-actualization,” “consciousness-raising,” and related expressions of self-concern are directly attributable to Hegel and the neo-Hegelian idealists. The dialectical triad of thesis, antithesis, and synthesis, which Hegel advanced as laws of thought expressed in logic, has now become a permanent fixture in the lexicon of undergraduate students and news commentators alike. In a subtler way, the triad found its way into Freudian theory:

Hegel -
God setup universe as "experiments" - using thesis anti- &
synthesis to find knowledge.

the “id” asserting that most fundamental of all theses; the “superego” standing in antithetical regard to it; and the “ego” emerging, synthetically, from the reconciliation of these counterpoised forces. Hegelianism, in its triumphant form, emerged in the middle years of the nineteenth century as Romanticism. In his *Reason in History*, Hegel proclaimed that “nothing great in the world has been accomplished without passion,”⁵⁹ and he argued that the essential nature of man—not one at a time but as a collective of consciousness—is freedom, the irrepressible freedom of spirit. Renewing the ageless dichotomy, he dismissed matter as the passive victim of natural laws and asserted spirit as the only free force in the universe. He and Beethoven were born in the same year. One set Goethe to music, the other to philosophy. In all, we find the romantic expression of Condorcet’s idea of progress. Escalated to the level of social action, we find a revolutionary spirit that has hardly begun to exhaust itself.

Karl Marx (1818–1883)

The historian of ideas is sorely tempted to discuss the roots of Marxist philosophy in a way that tends to trivialize the content of the philosophy. The temptation does not betoken or at least need not betoken enmity toward the philosophy or its author. Rather, from a purely philosophical perspective, so much of Marx’s thought is derivative that a mere enumeration of Marxist principles obscures the brilliant originality of his work taken as a whole.

Offsetting this tendency toward underestimation is an error in the opposite direction. Marx has proved to be such a powerful figure in the social and political affairs of our own century that it becomes tempting to read too much depth and genius into the exclusively philosophical aspects of his writings.

For our present purposes, we seek to do no violence to the stature of the man in social and political history by observing the very marginal influence of his speculations on the evolution of modern psychology. Marx did not have a great effect either on contemporaries who figured centrally in psychology or on later psychologists who, in fact, shared his materialistic orientation. It is true that in the highly politicized decades of Soviet science there would be eager attempts to establish the connection between Marxist theory and Pavlovian psychology but these attempts seldom rose higher than the level of mere propaganda and were never taken seriously by scientists of enduring consequence.

Marx’s failure to influence the course of psychological scholarship can be understood in a variety of ways. The nineteenth-century developments in psychology were of a largely experimental nature. Marx’s approach was historiographical and, in the loose sense, logical, and this was the very approach that the founders of experimental psychology were rejecting. Moreover, while Marx accorded “consciousness” a central role in his theory, it was a role that, on first inspection, was indistinguishable from Hegelianism and therefore not

We are driven by hist + economics (not psych).
All about competing for resources.
→ = power

likely to be serviceable. But even more than these dissonances was the unbreachable separation between psychology's commitment to the study of the individual and Marx's undivided attention to broad social processes. In short, Marx, was a sociologist at a time when psychology was being founded along biological lines. He discovered the psychology of alienation a century before social psychology would be prepared to study it. The same may be said of his recognition of those problems we now locate in such fields as "urban psychology," "industrial relations," and "community psychology." He saw the effects of industrialization on the family, on the worker, and on the relations among states. In these effects he perceived economic forces as the engine of all social and cultural and intellectual evolution.

We have noted Marx's acknowledged debt to Hegel. It is also noteworthy that his doctoral dissertation analyzed the systems of Democritus and Epicurus, a dissertation that could only have steeped him in the Enlightenment scholarship of Diderot and Condorcet. Indeed, in *The German Ideology* (1845–1846), where he discusses the successive stages of economic evolution in terms of tribal, state, feudal, and private ownership, we find almost a paraphrasing of Condorcet's *Sketch*. But unlike Condorcet or, for that matter, Epicurus, Marx was unwilling to place rational forces at the core of such evolution. For Marx, the materialism of consequence was historic materialism. He had no doubts but that man is so constituted biologically as to require and conform to historic materialism, but Marx was not to digress into anatomical or physiological reflections about "Man" while a world of men were suffering under the yolk of industrialism. Nevertheless, he was aware of the direct connection between the psychological materialism of the Enlightenment and the philosophical justifications of communism. This awareness is almost glibly rendered in *The Holy Family*:

As Cartesian materialism merges into natural science proper, the other branch of French materialism leads direct to socialism and communism. There is no need of any great penetration to see from the teaching of materialism on the original goodness and equal intellectual endowment of men, the omnipotence of experience, habit and education, and the influence of environment on man, the great significance of industry, the justification of enjoyment, etc., how necessarily materialism is connected with communism and socialism.⁶⁰

It is in the same work that Marx carries the analysis to its logical terminus. Since man is formed completely by his environment, by the social forces imposed upon him throughout his development, he cannot reasonably be held responsible for his crimes. These are the result of social evils and society is to blame. Consciousness itself is the creation of society.

Perhaps it is too trite to observe that a fair fraction of the common-sense psychologies of history can be partitioned in terms of attention to similarities

among men versus attention to differences. With a broad brush, we can color with the same hues egalitarianism, behaviorism, Marxism, socialism; with another hue, elitism, idealism, capitalism. Marx, like the empiricists he so admired, was persuaded that the similarities among men were far greater than their diverse stations would suggest. Noting how the steam engine alone had transformed the very character of the English nation, he was convinced that economic systems of production had imposed artificial class-differences upon the human community and that these differences must be eradicated. In retrospect, we tend to dismiss much of this as a kind of "folk" psychology, realizing that individual differences among people are not trivial and judging also that the psychological character of our race appears to survive a remarkable range of social and economic systems. Although we have hardly exhausted the set of important cross-cultural studies of cognitive and perceptual processes, there would seem to be enough data now to cast doubt upon such cornerstones of Marxist psychology as, "The nature of individuals thus depends on the material conditions determining their production."⁶¹ But the issue here is not whether Marx was "right" or "wrong" any more than it is whether his century was "right" or "wrong." Instead, we are to discover in *Marxism* that peculiar and fascinating theme uniting all the *isms* of the nineteenth century: positivism, materialism, utilitarianism, Hegelianism, pragmatism, and experimentalism. And what unites these otherwise immiscible movements is the confident belief that the world or the human enterprise or the heavens or everything can finally be encompassed by a grand vision validated by a faultless method. Scholarship, neither before nor since, is displayed with such certainty and finality. In the major works of physics, political theory, psychology, biology, sociology, and even more philosophy we discover an intellectual stridency and sureness that only amazes the twentieth century witness. In yet another way, this legacy of confidence also frustrates the modern citizen who is unable to locate the year when things started to go wrong; the time when certain physics became uncertain, when the knowable mind sunk once more to its historically unreachable depths, when the ping-pong rhythm of social organization gave way to cacophony.

There is yet another element in Marxist writing that warrants comment, although it, too, has no direct bearing on the evolution of modern psychology proper. The element is that of enmity and contempt, an element last present during the late Renaissance and Reformation. The wit and charm of the Enlightenment are absent. The canons of civility are ignored. Logic and compassion lose their struggle with impatience.

Nietzsche and Irrationalism

Little is served by attempts to analyze the "psychology" of intellectual leaders. Their importance is grounded in their ideas, not in their motives or personal

permanence requires human condition + stands up to it
is unattainable in reality

idiosyncracies. However, in Marx and in Nietzsche we confront an impatience and even a contempt never to be seen in the works of a Spencer or Bain or Mill. It will not do to propose that British philosophers were of a uniformly even disposition. Rather, what is at work is the passionate romanticism of Hegel: the Continental acceptance of emotion as a proper corollary of analysis. In observing this we call attention to the essential conservatism of nineteenth-century scholarship in Britain and France and the radicalism of thought flourishing in the German-speaking world. Now, it could only be a cliché to let the matter rest here. Of course, Marx, Nietzsche, and their disciples were radicals, but the utilitarians were also radicals in their ambitions. The difference is subtler. **Philosophy in Germany was speaking directly to the people. Philosophers were applying the ideas and the recognized forms of philosophical discourse to ends that were clearly political.** In their success, they introduced to scholarship a relevance that would have made even the *philosophes* uncomfortable.

Toward the conclusion of Part I of his *Beyond Good and Evil* (1886), Friedrich Nietzsche (1844–1900) proclaimed that **once we have sailed over and past the conventional morality of the philosophers, “psychology shall again be recognized as the queen of the sciences, to serve and prepare for which the other sciences exist. For psychology is now once again the road to the fundamental problems.”**⁶² Walter Kaufmann would come to call Nietzsche the first great “depth psychologist” owing to **Nietzsche’s identification of unconscious processes as the source of much of what we take to be the affairs of daily life and of culture itself.**⁶³ It was Nietzsche who traced the common features of all traditional philosophies to “the common philosophy of grammar,” and who anticipated much of today’s deconstructionist projects and Wittgensteinian analyses. In Nietzsche is to be found the most developed recognition of the cultural and linguistic sources of science, philosophy, and morality. Noting Boscovich’s pioneering insights in the eighteenth century, which reduced the putatively material atoms of the Newtonian world to nonmaterial centers of force, **Nietzsche celebrated the elimination of “substances,” “intuitions,” and other spiritous features of the traditional ontology.** Pressing on to levels that are pre-linguistic and primitive, he found as the veritable wellspring of culture that will to power previously analyzed by Schopenhauer. For Nietzsche, **the will to power is more basic than Darwinian survival instincts.** But, as with instincts, it operates beyond consciousness. It uses consciousness, with the latter being a merely superficial aspect of the mental. If the fundamental drive in life is not survival per se but the expression of power, then there are things that must always count more than life itself, not to mention more than the simple pleasures of the hedonist. **The power in question is the strength that is found not in the brute or bully, but in the capacity to endure and triumph over life’s menial obstacles.** His hero was Goethe, and the power in Goethe is at once the power of clarity

and of courage. Goethe knew there was more to life than the material composition of the Newtonian heavens or the Lockean mind; that life required a kind of devotion and sincerity ultimately expressive of levels of reality beyond mere sense and prosaic thought. It was Goethe, whether in his *Elective Affinities* or his theory of color, who relentlessly brought thought back into the thinker, value back into the world that is known. It was Goethe whose magisterial flights of artistry left in the distance the rubble of failed theories, each of them a prison of the mind. Goethe fulfilled the universal impulse: *He exercised his power.*

Then, too, there was the notorious friendship and falling out with Richard Wagner. Nietzsche was captured by the power of Wagner's music and its deliberate break with tradition; its contempt toward sentimentality; its deep roots in the most primal traditions of German culture; its purity of purpose. The letters between them disclose an unedited passion for social and moral theories, for theories of art and religion, and for theories of "types." As his own mind became a casualty of madness, Nietzsche's prose moved from the biting and perceptive to the dismal and merely peevish. The break with Wagner was occasioned in part by what Nietzsche took to be a concession to the faithful many, a toying with Christian superstitions. Like Marx, and unlike William James, he saw in religion a reflection of weakness and self-deception. Had he really been "the first great [depth] psychologist," he might have found in the universality of religious inclinations a vein of interests running deeper than the will to power.

In his celebrated (though rash and unconvincing) critique of Kant's metaphysics, Nietzsche found Kant charmed by the discovery of the pure categories and the possibility of synthetic knowledge a priori. As he understood Kant, all this is possible owing to a "faculty" of reason: So then we are able to do this because we have something that enables us to do this—the old *virtus dormitiva* of Moliere again. It is doubtful that any serious examination of Kantian metaphysics would result in so lighthearted a dismissal, but what is worth noting here is the conclusion Nietzsche reached, having satisfied himself of Kant's philosophical innocence (!). The conclusion reached is *not that we do not have such fixed features of thought and judgment, but that our having them does not establish their validity, only their utility.* We think and reason as we do in the furtherance of desires that are all too human. The discoveries of philosophy are but inventions, not the illumination of what is in the natural world or the world of independently existing entities, but the crafting of things expressive of cultural or personal value.

Marx and Nietzsche promised more than they delivered. The first was striving for a scientific understanding of events too utterly under-determined to admit of strict analysis. Nietzsche would put an end to the search of any certainties beyond the ambit of a given culture and its prejudices. He is not at all to

blame for what would take place in Germany a few decades after he declared that,

It is important that as few people as possible should think about morality—consequently it is *very* important that morality should not only one day become interesting!⁶⁴

Résumé

In this hurried survey of nineteenth-century psychological thought, the broad themes and special methods that would come to define modern psychology are all quite apparent. To this extent, the present chapter acknowledges the place of intellectual history in the evolution of modern psychology, and it is in this respect that the latter is a footnote, it were, to the recent past. But modern psychology, if not more than this recent past, is certainly different from it. Indeed, there is very little in our contemporary texts or journals that explicitly acknowledges a debt to the nineteenth century, and even less that sets out to settle the problems that so arrested past thinkers. What we need, then, is yet another bridge, one that might take us from the larger concerns and general counsel of the past to the narrower issues and more developed methods of the present. One bridge brought us from a philosophical to an allegedly scientific psychology, but not to genuinely modern psychology. In the next chapter, therefore, we will review more closely some of the developments already discussed and still other developments that succeeded in creating those perspectives, methods, and special schools that give color and contour to today's psychology.

Notes

1. No complete list can ignore Pierre Bayle (1647–1706), whose *Dictionnaire historique et critique* animated the scholarship of the Enlightenment with skeptical wit and an irreverent commitment to expose the pronouncements of authority to the twin lights of reason and evidence. Bayle's ten-volume *Dictionary* and related writings made him a figure of great consequence to such illustrious contemporaries as Locke and Leibniz. Later, he would be hailed by Voltaire and would, as well, influence the thinking of scholars as different in outlook as Berkeley and Hume. Except for his summary dismissal of distinctions between "primary" and "secondary" (Lockean) qualities, however, Bayle provides little material for psychological analyses. His immediate followers in philosophy were inspired by his courageous and telling assaults on the chained mind, but the bulk of his writings remain only peripheral to matters of central concern to psychology.

2. Voltaire, *Philosophical Letters*, translated by Ernest Dilworth, Bobbs-Merrill, Indianapolis, 1961.

3. Claude-Adrien Helvetius, *A Treatise on Man: His Intellectual Faculties and His Education*, translated by William Hooper, London, 1777.

4. Denis Diderot, *D'Alembert's Dream*, in *Diderot's Selected Writings*, edited by Lester G. Crocker and translated by Derek Coltman, Macmillan, New York, 1966, pp. 179–222.

5. Antoine-Nicolas De Condorcet, *Sketch for a Historical Picture of the Progress of the Human Mind* (1795), translated by June Barraclough, with an introduction by Stuart Hampshire, Noonday Press, New York, 1955.

6. Voltaire, *On the Pensées of M. Pascal*, in *Philosophical Letters*, p. 144.

7. In his final years, Kant had undertaken to oppose this transcendentalism. He died before completing what was to be a stern rebuke of those who would defend Romantic Idealism on Kantian grounds.

8. J. S. Mill, *Autobiography*. The edited volume by F. E. Mineka (Toronto, 1963) is especially useful.

9. J. S. Mill, *A System of Logic, Ratiocinative and Inductive: Being a Connected View of the Principles of Evidence and the Methods of Scientific Investigation*, Longmans, Green, London, 1900.

10. *Ibid.*, Book VI, Ch. III, Sec. I.

11. *Ibid.*, Sec. 2.

12. *Ibid.*, Ch. IV, Sec. 3.

13. *Ibid.*

14. *Ibid.*, Sec. 4.

15. *Ibid.*, Ch. 5.

16. *Ibid.*, Ch. V, Sec. 4.

17. J. S. Mill, *Utilitarianism*, in *The Utilitarians*, Dolphin Books, Doubleday, Garden City, N.Y., 1961, p. 404.

18. *Ibid.*

19. *Ibid.*, pp. 408–209.

20. *Ibid.*, pp. 432–433.

21. J. S. Mill spoke not only with the most eloquent voice of his age but, in major respects, with one most accurately reflecting the tone of that age. His contemporary, the essayist John Morley, observed in his eulogy for Mill, "Much will one day have to be said as to the precise value of Mr. Mill's philosophical principles. . . . However this trial may go, we shall at any rate be sure that with his reputation will stand or fall the intellectual repute of a whole generation of his countrymen." (*On the Death of Mr. Mill*, in *Nineteenth Century Essays*, University of Chicago Press, 1970.)

22. L. Levy-Bruhl, *The Philosophy of Auguste Comte*, translated by Frederic Harrison and published in English by Swan Sonnenschein, London, 1903. This was a most important edition, bringing the ideas of Comte to the English-speaking world. In America, in the 1850s, Henry Edger did much to advance the cause of positivism, but it was not really until Levy-Bruhl's careful study of the positivist program that a large number of philosophers in England and America seriously approached the system. On the earliest American forms, see *Positivism in the United States—1853–1861*, by Richmond, Laurin Hawkins, Harvard University Press, Cambridge, 1938.

23. Levy-Bruhl, pp. 191–193.

24. Antoine-Nicolas De Condorcet, *Sketch*.

25. Thomas Malthus, *An Essay on the Principle of Population as It Affects the Future Improvement of Society, with Remarks on the Speculations of Mr. Godwin, M. Condor-*

cet, and Other Writers. The essay has been reproduced in a paperback edition by the University of Michigan Press, Ann Arbor, 1959, with an introduction by Kenneth Boulding. The original appeared in 1798, in London.

26. Charles Darwin, *The Expression of the Emotions in Man and Animals*, Appleton-Century-Crofts, New York, 1896.

27. It was Ernst Haeckel (1834–1919) who formalized this Darwinian proposition into the “biogenetic law” according to which ontogeny recapitulates phylogeny. But Henri Bergson (1858–1941), almost single-handedly, began to lead an intellectual movement against Darwin’s materialistic theory of evolution. His *L’Évolution créatrice* (“Creative Evolution”), which was published in 1907, challenged evolutionary biology in terms very much like those used by the Gestalt psychologists in their arguments with associationists. Bergson found little plausibility in the notion that random processes could produce, in piecemeal fashion, such enormously complicated and functionally interdependent systems as those underlying vision, for example. Only a *creative* evolution, on Bergson’s account, could lead to the degree of biological organization of which the higher species are evidence. This creative evolution is powered by an *élan original*, a divine agency that has authored free will as well. That Bergson’s influence on contemporary psychology and especially contemporary American psychology has been less than commanding is to be understood in terms of the failure of such notions to be reduced to experimental modes of verification. Thus, we find Bergsonian ideas most prominently and perhaps uncomfortably located in that wide-ranging literature described as “existential psychology,” “humanistic psychology,” and “Gestalt,” psychotherapy. Perhaps their most cogent integration is to be found in the hypotheses advanced by Professor Jean Piaget, whose discussions of “cognitive development” are richly shaded by notions akin to *creative evolution*.

28. Francis Galton, *Hereditary Genius*, Macmillan, London, 1869.

29. The letter by Alexander Bain appears in R. M. Young’s recent and excellent study, *Mind, Brain, and Adaptation in the Nineteenth Century*, Oxford University Press, Clarendon, 1970, pp. 102–103.

30. As Professor Young notes, Bain’s *The Senses and Intellect* (1855) and *The Emotions and the Will* (1859) comprised the two-volume set that served as the standard British psychology for nearly fifty years. Both were published in London by the Parker Publishing Co. These volumes are discussed and reprinted without abridgement in Series A, Vols. IV and V, of *Significant Contributions to the History of Psychology*, D. N. Robinson, University Publications of America, Washington, D.C., 1978.

31. Johann, G. Spurzheim, *The Physiognomical System of Drs. Gall and Spurzheim*, Baldwin, Cradock, and Joy, London, 1815.

32. François Joseph Gall, *On the Functions of the Brain and of Each of Its Parts*, etc., 6 vols. translated by Winslow Lewis, Marsh, Capen, and Lyon, Boston, 1835.

33. Gall, *Functions*.

34. Herbert Spencer, *The Principles of Psychology*, Appleton-Century-Crofts, New York, 1896, p. 573.

35. *Ibid.*, p. 141.

36. Wilhelm Wundt, *Principles of Physiological Psychology*, Vol. I, translated by E. B. Titchener (from the fifth German edition of 1902), Macmillan, New York, 1904.

37. Gustav Fechner, *Elements of Psychophysics*, translated by Helmut Adler, edited

by David H. Howes and Edwin G. Boring, Holt, Rinehart and Winston, New York, 1966. The *Elements* was originally published in 1860.

38. Wundt, *Principles*, pp. 6–7.

39. Johann Friedrich Herbart (1776–1841) was a student of Fichte's. He held the chair once occupied by Kant at Königsberg. Unlike Fichte, he envisaged a scientific psychology based on mathematical analyses, a view he advanced in his *Psychologie als Wissenschaft* (1824–1825). Herbart's scientific psychology provided no room for innate ideas nor for a priori concepts. Rather, the elements of unconsciousness, including feelings, are to be understood in terms of the dynamic, mechanical laws of physics and are, therefore, mathematically expressible and deducible. He thus anticipated Fechner's psychophysical science and, in the Preface to the *Elemente*, Fechner acknowledges the debt: "To Herbart will always belong the credit not only of having been the first to point out the possibility of a mathematical treatment . . . but also of having made the first ingenious attempt to carry out such an enterprise; and everyone since Herbart will in this respect have to be second" (Fechner, *Elements of Psychophysics*). Davis H. Howes, and Edwin G. Boring,

40. Wundt, *Principles*, pp. 9–10.

41. *Ibid.*, p. 17.

42. Wilhelm Wundt, *Principles*, p. 193.

43. Marshall Hall, *Memoirs on the Nervous System* (1837; p. 70), in *Significant Contributions to the History of Psychology*, Series E, Vol. I, edited by D. N. Robinson, Greenwood Publishing, Connecticut, 1978.

44. *Ibid.*, pp. 70–71.

45. *Ibid.*, p. 74.

46. *Ibid.*, p. 110.

47. Wundt, *Principles*, pp. 8–9.

48. To appreciate in a historical way the approach of Fichte, it is useful to consult expositions of his works written at the time he began to enjoy wide attention outside Germany. Particularly penetrating in this respect is C. C. Everett's *Fichte's Science of Knowledge* (Chicago, S. C. Griggs and Company, 1892). In the same regard, consult E. B. Talbot's *The Fundamental Principles of Fichte's Philosophy*, New York, 1906.

49. W. T. Stace, *The Philosophy of Hegel: A Systematic Exposition*, originally published by Macmillan, 1924. This authoritative analysis of Hegel's *Encyclopaedia* and his *Phenomenology of Mind* is now available from Dover Publications, New York, 1955.

50. Bertrand Russell, *A History of Western Philosophy*, Simon and Schuster, Clarion paperback edition, New York, p. 730.

51. *Ibid.*

52. This remark appears in Karl Marx's Preface to the second edition of *Capital*. Marx moved away from orthodox Hegelianism in later years, but for a time he was as much a product of the Hegelian view of history and philosophy as anyone in the history of the movement. Only later did he and Engels "stand Hegel on his head."

53. Stace, *Hegel*, Sec. 75.

54. English editions of Hegel's *Encyclopaedia* are available. The William Wallace translation (Oxford University Press, 1873) can be found in later editions.

55. The authoritative translation of *Phenomenology of Mind* remains that by J. B. Baillie, London, 1910; 1931.

56. Stace, *Hegel*, p. 328.

57. *Ibid.*, p. 343.

58. The contemporary system of psychology bearing resemblance to Hegelianism is, of course, the cognitive psychology of Prof. Jean Piaget. The Piagetian stages of cognitive development begin with *egocentrism* and culminate with the ability to identify the connection between particular instances and universal propositions. Indeed, the six-stage evolution of cognition described by Piaget fits neatly into the Hegelian three stages of sensuous consciousness, sense-perception, and intellect. Piaget's psychology, often defined as an *evolutionary epistemology*, has strived to document by experimental demonstration that logical structure of thought which Hegel deduced. It is not surprising then, that contemporary challenges directed against Piagetian psychology are of a form almost indistinguishable from the British empiricists' attacks on Hegelianism.

59. G. W. F. Hegel, *Reason in History: A General Introduction to the Philosophy of History*, translated by Robert S. Hartman, Bobbs-Merrill, Indianapolis, 1953, p. 29.

60. *Karl Marx: The Essential Writings*, edited by F. L. Bender, Harper and Row, New York, 1972, pp. 145-146.

61. Karl Marx and Friedrich Engels, *The German Ideology*, Part 2, edited by C. J. Arthur, International Publishers, New York, 1976, p. 42.

62. Friedrich Nietzsche, *Beyond Good and Evil: A Prelude to a Philosophy of the Future*, translated by R. J. Hollingdale, Penguin Books, Hammondsworth, 1973, p. 36.

63. Walter Kaufmann, "Nietzsche as the First Great (Depth) Psychologist," in *A Century of Psychology as Science*, edited by Sigmund Koch and David Leary, McGraw-Hill, New York, 1985, pp. 91ff.

64. Nietzsche, *Beyond Good and Evil*, p. 138.

(Anthropo./evolut.)

Platited humans not completely diff from animals
- could study animals to help understand people

emp → materialism

scientific method → neurology/biology

Evolutionary ideas applied to all subjects, including psych-mental

11 From Systems to Specialties: The Crucial Half Century (1870–1920)

If we quickly thumb through the preceding chapters, one characteristic amid the diversity of topics and claims stands out and serves to divide all of the past from what is now taken to be the proper study of psychology. That characteristic is easier to detect than to define. Perhaps the term “system” serves most economically. If we ask, for example, what is common to the psychologies of Plato, Aristotle, Aquinas, Hobbes, Descartes, Locke, and the rest, it turns out to be their uniform commitment to develop a *system* of psychology able to embrace the fullest range of psychological phenomena: thought, emotion, memory, conduct, morality, government, aesthetics, and so forth. When we compare this commitment with the activities that most clearly identify contemporary psychology, we find a quite striking difference. To a certain degree, Freudian theory has been applied to a variety of personal and cultural matters and, to a lesser degree, modern behaviorism has attempted to address various aspects of social life and social organization. But **neither Freud nor the modern behaviorist would claim that his approach to psychology is designed to cover as broad a spectrum of phenomena as the approach that was routinely considered just a century ago.**

The more obvious explanations of this difference are uninforming. It could be argued, for example, that since earlier attempts failed, modern psychologists have learned to be more modest in their goals, more conservative in their speculations. Yet, if we look at the history of any significant intellectual undertaking, the same failures are common, but the surrender is not. Since the time of the pre-Socratics, there have been numerous attempts to account for all physical phenomena through a small set of universal laws, and none of these attempts succeeded. Nonetheless, today's theoretical physicist is just as committed as the ancients to an all-encompassing theory of matter. Similarly, the pages of intellectual history are filled with attempts to provide a faultless theory of politics, but none has surfaced and won the unopposed approval of the hu-